

CRM Lite — Enterprise CRM & AI Deal Intelligence

Complete Product Capabilities, AI Systems & Feature Specification Guide

A full-stack, enterprise-grade CRM engineered to streamline modern sales pipelines, automate follow-up governance, execute atomic customer conversions, deliver multimodal AI speech-to-text call analysis, provide a live-grounded CRM Copilot assistant, and deliver real-time KPI intelligence.

Document Ref:	CRMLITE-FEATURE-SPEC-2026	Release Version:	v2.2.0 Production Release
Frontend Framework:	React 19 / Vite 8.3 / React Router v7	UI Design System:	Vanilla CSS Dark Glassmorphism
Backend Engine:	Python 3.14 / Django 6.1 / DRF 3.18	Database Systems:	PostgreSQL (Production) / SQLite (Dev)
AI Infrastructure:	Google GenAI / Gemini 2.5 Flash	AI Capabilities:	Speech-to-Text, Copilot, Dossiers
Microservice:	FastAPI Federated Global Search	Security / Auth:	JWT Bearer Auth + 3-Tier RBAC

Pre-Configured Demo User Credentials (1-Click Login Enabled)

Role Persona	Email / Username	Password	Access Tier & System Permissions
Admin / Mentor	admin@crmlite.com	Admin@123	Full root control, user provisioning, lead deletion, global analytics
Sales Manager	manager@crmlite.com	Manager@123	Team pipeline visibility, lead assignment, customer conversion, CSV reports
Sales Executive (Alex)	alex@crmlite.com	Alex@123	Assigned leads, follow-up scheduler, call notes, personal metrics
Sales Executive (Sarah)	sarah@crmlite.com	Sarah@123	Assigned leads, follow-up scheduler, call notes, personal metrics

SYSTEM SCOPE & INTELLECTUAL PROPERTY NOTICE

This document provides the authoritative, exhaustive technical and functional specification for CRM Lite. It catalogs all user-facing capabilities, artificial intelligence pipelines, REST endpoints, data validations, and security protocols active in the production codebase.

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1. Executive Summary & Complete Feature Ecosystem

CRM Lite is a unified sales operations and lead management platform engineered for fast-moving enterprise sales teams. The system eliminates manual sales friction, accelerates deal velocity, enforces operational rigor, and unlocks generative AI capabilities across customer interactions.

Unlike legacy CRM platforms burdened with bloated plugins and prohibitive licensing costs, CRM Lite delivers a responsive, modern single-page application (React 19) backed by high-throughput Django REST Framework endpoints and an optional FastAPI federated search engine. Every feature—from atomic customer conversions to multimodal call transcriptions—is built with transactional safety, rigorous access controls, and zero hardcoded metrics.

Master CRM Feature Ecosystem Matrix

Module / Domain	Key Capabilities	Primary Benefit
Lead Management & 360° View	Full lifecycle tracking, priority scoring, multi-channel sourcing, duplicate warning, loss reason attribution, chronological activity audit trail.	Complete visibility into prospect history and eliminated lead attrition.
Visual Sales Pipeline	7-stage interactive Kanban board, real-time stage currency valuation, deal counter, drag-and-drop movement, rep/priority filtering.	Clear operational funnel forecasting and accelerated deal progression.
Smart Follow-Ups	Automated overdue detection engine, anti-past scheduling validation, 'Due Today' hub, outcome completion logging.	Zero missed follow-up deadlines; strict rep accountability.
Sales Calendar	Month/Week/Day scheduling layouts, color-coded task cards, direct event booking, quick completion modals.	Holistic time management and seamless appointment tracking.
Customer Accounts	Atomic lead conversion inside transaction.atomic(), linked original lead lineage, Lifetime Value (LTV) metrics.	Clean separation between prospect pipeline and paying accounts.
AI Call Deal Intelligence	In-browser audio recorder, multi-format audio upload (up to 25MB), Gemini 2.5 Flash native transcription, structured intent/objection extraction.	Automated CRM call note logging; hours of manual data entry saved.
Conversational CRM Copilot	Slide-out drawer chatbot grounded in live DB data with confirmation cards for safe write actions (lead creation, follow-ups).	Instant natural-language queries and guided sales workflows.
Team Collaboration	Internal private team notes, @username mentions autocomplete, real-time trigger notifications.	Seamless handoffs and transparent deal collaboration.
Notification Engine	In-app notification bell with unread badge + responsive HTML transactional emails for stage changes, assignments, and overdue.	Real-time awareness across reps and managers on critical events.
Real-Time Analytics	Zero hardcoded stats: win rates, pipeline value, monthly trends, rep performance scorecards, and streaming CSV downloads.	Executive-level visibility into conversion velocity and rep ROI.

2. Feature 1: 360° Lead Management & Dossier

Lead management represents the foundational core of CRM Lite. Every prospect is represented by an exhaustive data record capturing contact coordinates, commercial attributes, interaction histories, and predictive indicators.

2.1. Lead Attributes & Schema Specifications

Each lead record incorporates the following normalized attributes:

Identity & Contact	Full Name, Corporate Email Address, Direct Phone / Mobile Number, Organization / Company Name, Job Title, Industry Sector.
Pipeline Metrics	Estimated Deal Value (Numeric Currency), Win Probability % (0–100 slider), Expected Close Date, Priority Tier (LOW, MEDIUM, HIGH, URGENT).
Source Attribution	Ingestion channel attribution: WEBSITE, LINKEDIN, REFERRAL, COLD_OUTREACH, TRADE_SHOW, WEBINAR, OTHER.
Ownership & Status	Current Pipeline Stage (NEW through WON/LOST), Assigned Representative (User Foreign Key), Lead Qualification Score (0–100 integer).
Audit Metadata	Created By User, Created Timestamp, Last Updated Timestamp, Handover / Assignment Timestamp.

2.2. Duplicate Lead Warning Engine

To prevent duplicate outreach and sales rep collisions, the system monitors incoming lead creation requests. If a prospective lead matches an existing email address or phone number, the frontend surfaces an intelligent duplicate warning banner linking directly to the existing lead record, while the backend creates a DUPLICATE_WARNING notification for management review.

2.3. Multi-Channel Interaction Logging & Activity Timeline

Every interaction with a prospect is chronologically logged into the immutable ActivityLog timeline. Sales reps can categorize interactions using structured types:

Channel Type	Use Case	Timeline Representation
CALL	Inbound or outbound telephone conversation	Phone receiver badge, call duration, summary
WHATSAPP	Direct instant messaging exchange	Chat bubble badge, message excerpt, next response date
EMAIL	Written correspondence or quote dispatch	Mail envelope badge, subject line, deliverability
MEETING	Face-to-face or video conference conference	Calendar badge, attendee list, meeting minutes
DEMO	Formal solution demonstration or proof-of-concept	Monitor badge, feature feedback, technical objections
OBJECTION	Specific prospect reservation (budget, timing, competitor)	Warning shield badge, objection category, counter-strategy
GENERAL	Internal administrative note or desk research	Note page badge, research links, account updates

2.4. Mandatory Loss Reason Attribution

When a deal moves into the LOST terminal stage, CRM Lite displays the LostReasonModal. Sales reps cannot disqualify a deal without selecting a structured root-cause category (e.g., *Budget Constraints*, *Competitor Selection*, *Feature Gap*, *Poor Timing*, *Unresponsive Prospect*) and documenting a detailed post-mortem note. This data feeds directly into the executive win/loss analytics engine.

2.5. Atomic Customer Conversion Engine

When a deal is marked WON or reaches final agreement, Managers and Admins can convert the lead into an official Customer. This operation executes within a strict `django.db.transaction.atomic()` boundary to guarantee ACID database consistency:

TRANSACTIONAL CONVERSION PROTOCOL (`transaction.atomic()`)

1. Validates that the lead is not already converted (prevents double-conversion idempotency errors).
2. Spawns a new Customer entity with company details, lifetime value, and original lead foreign key.
3. Sets lead status to WON, sets `is_converted=True`, and records conversion timestamp.
4. Seamlessly re-links all historical activity logs, communication notes, and call recordings to the customer profile.
5. Dispatches in-app notification and email confirmation to the account owner.

3. Feature 2: Interactive Visual Sales Pipeline (Kanban)

The Visual Pipeline transforms linear database queries into an intuitive, multi-column Kanban board allowing sales reps and executives to grasp pipeline health at a glance. Every deal is rendered as a responsive card displaying prospect details, deal value, priority badges, next follow-up dates, and assigned rep avatars.

3.1. The 7-Stage Sales Lifecycle Funnel

Pipeline Stage	Key Stage Objective	Automated Action / Trigger
1. NEW	Raw incoming prospect awaiting initial qualification and rep assignment.	Auto-notifies sales manager of unassigned deal; prompts priority triage.
2. CONTACTED	First outreach attempted via telephone, LinkedIn, or personalized email.	Enforces follow-up scheduling within 48 hours to maintain momentum.
3. DEMO_SCHEDULED	Formal product walk-through or discovery session locked in calendar.	Syncs to Sales Calendar; sends calendar invite reminders.
4. NEGOTIATION	Commercial quote or contract presented; commercial pricing under review.	Flags deal as high-priority in dashboard forecast; tracks objection logs.
5. QUALIFIED	Budget confirmed, decision maker committed, final legal/security review.	Unlocks 1-click 'Convert to Customer' action for managers.
6. WON	Contract executed, payment initiated, commercial victory achieved.	Dispatches celebratory email alert; updates rep quota scorecards.
7. LOST	Prospect disqualified or chose competitor; terminal sales outcome.	Opens LostReasonModal to record mandatory attribution data.

3.2. Real-Time Column Value & Deal Aggregation

Each stage column header dynamically calculates and displays two critical metrics in real time:

- **Deal Counter:** Total active prospects residing in this stage.
- **Aggregate Stage Currency:** Summed estimated deal value (e.g. \$148,500.00), updated instantly upon card movement.

3.3. Multi-Parameter Pipeline Filtering

Sales reps and managers can filter the entire Kanban board instantaneously without page reloads using:

- **Assigned Sales Rep:** Filter by specific team member or view unassigned leads.
- **Priority Tier:** Isolate URGENT and HIGH priority deals requiring immediate closing action.
- **Time Window:** Filter by leads created this week, this month, or this quarter.
- **Search Keyword:** Live fuzzy matching against prospect name, organization, or email.

4. Feature 3 & 4: Follow-Up Engine & Sales Calendar

In enterprise B2B sales, deal velocity is governed by rigorous follow-up cadence. CRM Lite replaces scattered calendar reminders with a centralized, rules-enforced Follow-Up Engine and Interactive Sales Calendar.

4.1. Strict Follow-Up Validation & Anti-Past Scheduling Rules

To ensure sales data integrity, both frontend forms and backend serializers enforce strict temporal validation. The system strictly prohibits scheduling follow-up appointments in the past: `scheduled_time >= timezone.now()`. Attempts to submit a past timestamp immediately trigger a descriptive 400 Bad Request error.

4.2. Automated Overdue Follow-Up Engine

The follow-up engine continuously evaluates open tasks against the current system time. Tasks where `scheduled_time < now()` and `status == 'PENDING'` are automatically elevated:

- **Visual Overdue Badging:** Marked with high-contrast red warning badges and elapsed time tags (e.g., *'3 days overdue'*).
- **Dashboard KPI Counter:** Increments the executive 'Overdue Tasks' alert widget.
- **Notification Trigger:** Sends in-app reminders to both the assigned sales rep and their sales manager.

4.3. 'Due Today' Action Hub & Completion Workflow

Upon logging in, sales reps are greeted with the 'Due Today' queue. Completing a follow-up requires structured accountability:

1. Rep clicks **'Complete Follow-up'**.
2. A modal prompts for mandatory **Outcome Notes** (what occurred during the interaction).
3. Rep records the interaction result (e.g., *'Client requested formal quote'* or *'Decision maker out of office'*).
4. The system prompts to optionally schedule the next follow-up appointment in one continuous flow.

4.4. Interactive Sales Calendar Hub

The Sales Calendar page (Calendar.jsx) provides a comprehensive scheduling cockpit featuring:

- **Multi-View Modes:** Seamlessly toggle between Month, Week, and Day layouts.
- **Color-Coded Event Cards:** Distinct palette identifying Calls (Teal), Meetings (Sky Blue), Demos (Purple), and Urgent Tasks (Rose).
- **Direct In-Calendar Booking:** Click any open time slot to open the scheduling modal with pre-filled datetime.
- **1-Click Record Navigation:** Click any event card to view the associated prospect dossier or mark the task complete.

Component	Key Functional Capability	Business Value
FollowUps.jsx	Filterable task table with tabs for Overdue, Today, Upcoming, and Completed tasks.	Ensures reps never lose track of a commitment or prospect touchpoint.
Calendar.jsx	Full visual calendar grid with drag navigation, date jump, and event density indicators.	Prevents double-booking and optimizes sales call scheduling.
CalendarScheduleModal	Integrated booking drawer linking lead records, contact methods, and priority flags.	Sub-minute scheduling workflow directly from calendar view.

5. Feature 5: Converted Customer Directory & Accounts

A critical architectural principle of CRM Lite is the strict operational boundary between *Sales Prospects* (unconverted leads) and *Paying Customers*. Once a lead is converted, they graduate from the transient sales pipeline into the persistent Customer Directory.

5.1. Customer 360° Profile & Account Record

- The Customer record (customers/models.py and CustomerDetails.jsx) preserves full institutional knowledge:
- **Primary Coordinates:** Company Name, Primary Contact Person, Direct Email, Phone, Office Address, Industry.
 - **Original Lead Lineage:** Unbroken foreign key link to the originating Lead entity, recording original source, initial inquiry date, and conversion duration.
 - **Lifetime Value (LTV) Metrics:** Tracks total closed contract value, ongoing subscription tier, and contract renewal dates.
 - **Preserved Interaction Archive:** Full access to historical pre-conversion notes, call audio summaries, objection logs, and emails.

5.2. Customer Account Directory Features

- The Customer Directory page (Customers.jsx) provides enterprise account management tools:
- **Instant Account Search:** Real-time filtering by company name, contact person, or corporate email.
 - **Account Status Filters:** Segment customers by Active, Onboarding, In-Renewal, or Past-Due.
 - **LTV Sorting:** Rank accounts by financial value to support executive customer success triage.

6. Feature 6: Cutting-Edge Artificial Intelligence (AI) Suite

CRM Lite implements a state-of-the-art enterprise AI subsystem powered by Google Gemini 2.5 Flash. Rather than superficial chatbot wrappers, the AI architecture is deeply integrated into operational workflows: transcribing audio sales calls, grounding conversational queries in live CRM data, and synthesizing executive dossiers.

DECOUPLED AI PROVIDER ARCHITECTURE (ai_provider.py)

The backend uses an abstract base provider (BaseAIProvider) pattern. All LLM calls pass through standardized contracts (transcribe_audio, summarize_call, chat). Google GenAI (gemini-2.5-flash) is the primary production implementation. The architecture allows swapping in Anthropic Claude, OpenAI, or local Ollama models by registering new providers in the factory without altering any view or controller logic.

6.1. Multimodal Speech-to-Text & Call Deal Intelligence

Eliminates hours of manual data entry by extracting verified deal intelligence directly from audio sales call recordings:

In-Browser Audio Recording	The AudioRecorder component captures real-time microphone streams via the browser navigator.mediaDevices.getUserMedia API in audio/webm format with active waveform visualizations and duration timers.
Multi-Format Audio Upload	Accepts pre-recorded sales call files (.mp3, .wav, .m4a, .webm, .ogg, .aac) up to 25MB with strict MIME-type and size validation in TranscriptionService.
Direct Multimodal Processing	Submits raw audio bytes directly to Gemini 2.5 Flash using types.Part.from_bytes(). Bypasses third-party speech-to-text intermediaries, dramatically reducing latency and transcription error rates.
Structured Deal Extraction	Guided by a strict system prompt and schema, the model outputs structured JSON with zero markdown hallucination containing: verbatim dialogue transcript, 2-sentence executive summary, customer intent, detected objections, overall sentiment, stage recommendations, and actionable next steps.
1-Click CRM Synchronization	From the call intelligence page, reps can link the call to an existing lead, update pipeline stage, and append extracted action items directly to the lead's activity timeline with a single click.

6.2. Conversational CRM AI Copilot (Slide-out Assistant Drawer)

Available from any screen via the floating sparkle button or navigation header, the AI Assistant Drawer (AiAssistantDrawer.jsx) delivers a live conversational copilot grounded directly in the user's CRM database.

Live Database Grounding: When a query is submitted, CRMTools extracts authenticated user context—active leads, assigned pipeline values, overdue follow-ups, and customer metrics—and injects this structured context into the system prompt. The model cannot invent fake leads or hallucinate deal numbers.

Safe Action Execution (Human-in-the-Loop Protocol): The Copilot can assist with write operations, but strictly adheres to a two-phase confirmation protocol:

COPILOT ACTION CONFIRMATION PROTOCOL

- Draft Phase:** When the user asks 'Schedule a follow-up with Acme Corp tomorrow at 2 PM', the AI returns a structured pending_action object specifying action type (schedule_followup, create_lead, update_stage) and payload parameters.
- Interactive UI Card:** The frontend renders an interactive confirmation card detailing the exact entity, date, and changes.
- Human Execution:** The database mutation is executed ONLY when the user clicks 'Confirm & Execute'. If rejected, the draft is discarded.

6.3. Lead Dossier AI Synthesis & Prescriptive Recommendations

From the Lead Details screen, sales reps can click '**Generate AI Briefing**'. The backend aggregates the prospect's entire timeline—every interaction note, stage transition, call summary, and follow-up outcome—and prompts Gemini 2.5 Flash (temperature 0.2) to synthesize an executive dossier containing:

- **Executive Synthesis:** A 3-sentence summary of the prospect's organizational needs and commercial relationship.
- **Win Probability Breakdown:** Data-driven factors impacting deal success.
- **Key Objections Matrix:** Unresolved technical, pricing, or organizational blockers.
- **Prescriptive Next Actions:** Highly specific tactical recommendations to advance the deal to the next pipeline stage.

6.4. Dynamic Environment Hot-Reloading

To maximize development velocity and prevent downtime, GeminiProvider._resolve_api_key() includes a dynamic environment reloader. If an administrator updates the Gemini API key in the Settings page or edits backend/.env, the AI provider detects the change on the fly without requiring a server reboot.

6.5. Central AI Operations Center (AiCenter.jsx)

The dedicated AI Center acts as the operational command dashboard for generative features. Sales leaders can inspect:

- **Live Provider Status:** Verifies connectivity with Google GenAI / Gemini 2.5 Flash.
- **Feature Launchers:** Direct 1-click launchpad for Call Deal Intelligence, Copilot Assistant Drawer, and Lead Synthesis.
- **Token & Usage Metrics:** Visual indicators tracking daily audio transcriptions and copilot query volumes.

6.6. AI Configuration & Model Tunables (Settings.jsx)

Administrators can adjust AI parameters directly from the UI Settings page:

- **Secure Key Storage:** Enter or rotate Google Gemini API keys with input masking.
- **Temperature Sliders:** Calibrate model creativity vs. strict grounding (0.2 for call summaries; 0.3 for chatbot).
- **Model Selection:** Switch between gemini-2.5-flash (low latency) and gemini-1.5-pro for deep analysis.

7. Feature 7: Team Collaboration & Internal Mentions

Complex enterprise deals require cross-functional teamwork between reps, sales engineers, and managers. CRM Lite includes a specialized Internal Comments & Mentions system (InternalCommentsSection.jsx and MentionInput.jsx).

- **Private Team Comments:** Distinguishes internal strategy remarks from official prospect communication logs. Internal comments are never visible to external contacts.
- **Dynamic @User Mention Autocomplete:** Typing @ in any comment box triggers an instant dropdown matching active sales team members.
- **Instant Alert Dispatch:** Tagging a colleague automatically dispatches an INTERNAL_MENTION in-app notification linking directly to the comment.
- **Handoff Auditability:** Enables sales managers to leave strategic instructions during deal reassignment.

8. Feature 8: Notification & Real-Time Alert Engine

CRM Lite features a dual-channel notification architecture: a real-time In-App Notification Center paired with an automated transactional HTML Email Dispatcher.

8.1. In-App Notification Center

Located in the top navigation bar (NotificationBell.jsx & NotificationDropdown.jsx):

- **Live Unread Badge:** Displays a counter of unread alerts, updating dynamically upon event dispatch.
- **Priority Color Indicators:** Categorized by priority tiers: LOW, NORMAL, HIGH, and URGENT.
- **1-Click Entity Navigation:** Clicking an alert marks it as read and immediately routes the user to the related Lead, Customer, or Follow-up.
- **Batch Management:** Includes 'Mark All as Read' and filtered views for unread vs. historical alerts.

8.2. Automated Transactional HTML Email System

Critical business events trigger high-deliverability transactional emails formatted using modern, responsive HTML templates (e.g., lead_stage_updated.html):

Business Event Trigger	Recipient	Email Template & Content
Lead Stage Transition	Assigned Sales Rep & Sales Manager	Branded HTML notification detailing old vs. new stage, deal value, and next recommended action.
Lead Assignment / Handover	Newly Assigned Sales Rep	Welcome briefing containing prospect contact info, lead source, priority, and previous rep notes.
Follow-up Overdue Alert	Assigned Rep & Escalated to Manager	Urgent reminder highlighting elapsed deadline, prospect value, and direct link to complete task.
Customer Converted	All Team Members & Management	Commercial victory alert celebrating closed deal, converted revenue, and customer onboarding link.

8.3. Anti-Self-Notification Filtering

To prevent notification noise, NotificationService enforces a strict anti-self-notification guard: if an action is performed by User A (e.g. User A updates a stage or marks a follow-up complete), the service automatically suppresses sending User A a notification for their own action.

8.4. Real Email SMTP Testing & Diagnostics (check_email.py)

To guarantee 100% email deliverability in production, the backend includes an automated diagnostic CLI tool:

- **Verification Command:** Run `python backend/check_email.py [recipient_email]`.
- **Pre-Flight Checks:** Verifies TLS port 587 connectivity, credentials syntax, and Google 16-character App Passwords.
- **Instant Test Dispatch:** Sends a verified end-to-end test message to confirm inbox delivery and spam filter bypass.

9. Feature 9: Real-Time KPI Dashboards & Analytics

A standout architectural achievement of CRM Lite is that **zero analytics metrics are hardcoded**. Every KPI card, conversion ratio, pipeline valuation, and monthly chart aggregates live database records in real time.

9.1. Executive KPI Dashboard Widgets

The Dashboard (Dashboard.jsx) renders six live metrics updating dynamically on data change:

- **Total Active Leads:** Total count of in-flight prospects currently progressing through the sales pipeline.
- **Converted Customers:** Total official customer accounts created through the atomic conversion engine.
- **Overall Win / Conversion Rate %:** Calculated dynamically as $(\text{Converted Leads} / \text{Total Finished Leads}) * 100$.
- **Estimated Pipeline Value:** Sum of estimated values across all open deals (formatted in international currency).
- **Overdue Follow-up Alert:** High-priority counter reflecting past-due tasks requiring immediate escalation.
- **Monthly Inflow Trend:** Time-series aggregation showing new prospects acquired month-over-month.

9.2. Sales Representative Performance Scorecard

The Reports module (Reports.jsx) provides granular rep-by-rep performance evaluation: total leads handled, deals won, deals lost, individual conversion percentages, and total closed contract value.

9.3. High-Performance Streaming CSV Export Engine

Managers and Admins can export filtered CRM records at any time. Rather than buffering large datasets in memory, the backend implements Django StreamingHttpResponse to stream CSV rows directly to the client browser, preventing memory spikes even with tens of thousands of records.

10. Feature 10: Security, RBAC & Authentication

CRM Lite enforces strict Role-Based Access Control (RBAC) across three distinct tiers. Security is enforced at the database query level (scoping QuerySets in DRF viewsets), controller permission classes, and client-side route guards (RoleGuard.jsx).

Feature / Operational Capability	Admin / Mentor	Sales Manager	Sales Executive
User Provisioning & Team Management	Full Access	View Only	Forbidden (403)
Lead Pipeline Visibility Scope	All Organization Leads	All Team Leads	Assigned / Created Only
Lead Creation & Interaction Notes	Allowed	Allowed	Allowed
Lead Assignment & Reassignment	Allowed	Allowed	Forbidden (403)
Customer Conversion Execution	Allowed	Allowed	Forbidden (403)
Hard Delete Lead Record	Allowed	Forbidden (403)	Forbidden (403)
View Organization Performance Analytics	Full Metrics	Team Metrics	Own Metrics Only
Export Financial & Lead Reports to CSV	Allowed	Allowed	Forbidden (403)
AI Call Intelligence & Recording Upload	Allowed	Allowed	Allowed
Conversational AI Copilot Drawer	Org-Scoped Grounding	Team-Scoped Grounding	Self-Scoped Grounding
Configure AI API Keys & System Settings	Full Access	Forbidden (403)	Forbidden (403)

10.1. Authentication Architecture (JWT Rotation)

Authentication is powered by `django-rest-framework-simplejwt`. Clients authenticate with an email and password to receive a short-lived access token and a long-lived refresh token. The frontend `Axios` interceptor transparently negotiates token rotation upon receiving a 401 response, preventing mid-session user logouts while safeguarding against token theft.

11. Feature 11: Architecture, APIs & Microservices

CRM Lite is engineered around a modern decoupled architecture. The React 19 Single Page Application communicates via REST APIs with the Django backend and optionally queries a dedicated FastAPI search microservice.

HTTP Method & Route	Controller / View	Functionality & Scope
POST /api/auth/token/	CustomTokenObtainPairView	Authenticate credentials; issue JWT pair
POST /api/auth/token/refresh/	TokenRefreshView	Renew expired access token using refresh token
GET/POST /api/leads/	LeadViewSet	List leads (role-filtered) and create new prospects
GET/PATCH /api/leads/:id/	LeadViewSet	Retrieve lead details or update attributes/stage
POST /api/leads/:id/convert/	LeadViewSet.convert	Execute atomic conversion to official Customer
POST /api/leads/:id/ai_summary/	LeadViewSet.ai_summary	Synthesize comprehensive AI lead dossier
GET/POST /api/follow-ups/	FollowUpViewSet	List and schedule follow-ups with anti-past validation
POST /api/follow-ups/:id/complete/	FollowUpViewSet.complete	Log outcome notes and mark task finished
POST /api/ai/call-summary/	AICallSummaryView	Multimodal audio upload and deal analysis
POST /api/ai/chat/	AIChatView	Conversational Copilot query with CRM grounding
POST /api/ai/execute-action/	AIActionExecuteView	Execute human-confirmed Copilot write action

GET /api/reports/dashboard/	ReportDashboardView	Zero-hardcoded real-time KPI aggregations
GET /api/reports/export-csv/	ReportExportCSVView	Streaming HTTP response generating lead CSV export
GET /api/fastapi/search/	FastAPI Service (:8001)	Ultra-low-latency federated prospect & note search

12. Feature 12: Modern UI/UX Design System

CRM Lite strictly avoids bulky, generic CSS utility frameworks. The frontend features a bespoke, hand-crafted Vanilla CSS design system engineered for instantaneous load times and rich visual aesthetics.

Dark Glassmorphism	Deep charcoal slate backgrounds (#0B0F19) paired with frosted glass cards using backdrop blur filters (backdrop-filter: blur(12px)) and subtle translucent border glows.
Modern Typography	Pairing Google Fonts Outfit (for bold, modern headings and metric displays) with Plus Jakarta Sans (for ultra-crisp body copy and tabular data).
Responsive Layouts	Fluid CSS Grid and Flexbox layouts designed for high-density desktop displays, standard laptops, and mobile tablet viewports.
Micro-Interactions	Subtle transition effects on button hovers, smooth modal slide-ins, and pulsating status badges for urgent/overdue states.
Zero Dependency Bloat	Zero Tailwind runtime bloat or fragile post-CSS build dependencies; lightning-fast Vite compilation.

13. Quality Assurance & Developer Cheatsheet

Automated Test Suite Verification

CRM Lite features automated test coverage across authentication, role permissions, past follow-up rejection, and atomic customer conversion transactions. Execute the suite via:

```
python backend/manage.py test accounts leads followups notifications
```

Quick Local Start Commands

Step	Command Line Instruction	Description
1. Backend	python backend/manage.py runserver 8000	Launches Django REST API at http://127.0.0.1:8000/api/
2. Frontend	cd frontend && npm run dev	Launches React 19 Vite SPA at http://localhost:5173/
3. Microservice	uvicorn fastapi_service.main:app --port 8001	Launches optional FastAPI search microservice
4. Seed Data	python backend/manage.py seed_data	Populates realistic leads, calls, notes, and metrics
5. Interactive Docs	Visit http://127.0.0.1:8000/api/docs/	OpenAPI 3.0 / Swagger UI interactive API console

SPECIFICATION APPROVAL & RELEASE ATTESTATION

This product and feature specification represents the verified state of the CRM Lite platform. All features detailed herein—including the 7-stage visual pipeline, atomic customer conversion, multimodal audio transcription, CRM copilot drawer, and real-time KPI dashboards—are fully functional, tested, and ready for production deployment.